Jacob Groner – ML Research Engineer

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Education

Cornell University, College of Arts & Sciences, Ithaca, NY — Graduated December 2025

BA in Computer Science and Mathematics, Minor in Biological Sciences

Machine Learning • AI Reasoning/Decision-Making • Reinforcement Learning • Program Synthesis • Object Oriented Programming • Discrete Structures • Data Structures and Functional Programming • Analysis of Algorithms • Linear Algebra • Multivariable Calculus • Statistics • Differential Equations • Complex Analysis

Selected Projects

MIRAGE – LLM enhancement to improve generalization

- Developed hybrid architecture combining transformer language model with Hidden Markov world model
- Extracted latent structural rules from training data to improve out-of-distribution reasoning and generalization

Quoridor Bot – Search-Based Game Agent

- Implemented Monte Carlo Tree Search agent for adversarial board game
- Integrated convolutional neural network evaluation function for early-cutoff search

Experience

NeuroAI Research, Lab of Professor Weinan Sun, Ithaca, NY — May 2024 – February 2026

- Created wavefront pathfinding algorithm with Hidden Markov Model
- Identified hippocampal place cells from Calcium imaging data
- Created virtual reality mazes for headfixed mice
- Created virtual reality mazes and trained mice to navigate them while headfixed

Neuroscience Research, Lab of Professor Ian Ellwood, Ithaca, NY — August 2021 – November 2022

- Built virtual reality system head fixed mouse navigation and brain imaging
- Completed wiring, mechanical construction, and software elements of Parker Motion products
- Programmed Unity maze representation and control system

Software Developer, Elminda, Tel Aviv, Israel — May 2022 – August 2022

- Designed LSTM noise detection model for multichannel EEG signal
- Improved tool for eye and muscle artifact labeling
- Performed data augmentation

Instructor, ID Tech, Irvine, CA — June 2021 – August 2021

- Taught computer science courses in Java, Python, AI, and Game Design
- Prepped for and taught a two-hour class every weekday
- Instructed groups of students ages 8 -17

Skills

Workflow Preferences: OpenAI Gym • PyTorch • Jax • Pandas • Numpy • Plotly • VSCode • Windows • Google Drive • Codex

ML Techniques: DQN • PPO • LSTM • Transformers • RAG • CNN • VAE • HMM • EM • A* • MCTS • PCA

Languages: English (native) • Hebrew (Conversational)

Hobbies

Chess • Bass Guitar • Weightlifting • Surfing • Ultimate Frisbee